

Alzheimer's Disease Detection Using Brain MRI Images



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Team Details

- Bibek Giri (LLM Research and Patient Management)
- Mahesh Rawat (Dataset Preparation, Database Integration and Grad-CAM)
- Prabhat Kumar Sardar (Model Training and Backend Integration)
- Yukesh Dura (Frontend Development and UI/UX Design)

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Introduction, Problem Statement & Objectives

Introduction:

- Alzheimer's Disease affects brain memory and cognitive functions.
- Early diagnosis improves treatment planning.
- MRI scans contain important structural brain information.
- Deep learning can automate MRI analysis.

Problem Statement:

- Manual MRI analysis requires expert neurologists.
- Early-stage symptoms are difficult to detect.
- Healthcare systems lack automated diagnostic tools.
- Misdiagnosis can delay treatment.

Objectives:

- To develop a 4-class MRI classification system.
- To compare multiple deep learning models.
- To implement Grad-CAM for explainability.

Proposed System Functionalities

MRI Processing:

- MRI Image Upload
- Image Preprocesses: Grayscale + Resize to 224x224
- Removes corrupted images permanently
- Removes duplicate images permanently

AI Prediction:

- Alzheimer's Stage Classification
- Confidence Score Generation
- Risk Level Detection

Explainable AI:

- Grad-CAM Heatmap
- Brain Region Highlighting
- Prediction Explanation

System Features:

- Flask Backend API
- Frontend Integration
- Dashboard System
- Database Storage
- Patient History Tracking
- LLM-based Report Generation

Completed Functionality

Dataset Preparation:

- Dataset Collection
- Dataset Cleaning
- Data Augmentation
- Dataset Balancing

Model Training:

- CNN Training
- MobileNetV2 Training
- DenseNet121 Evaluation
- ResNet50 Fine-Tuning

Evaluation:

- Model Comparison Completed
- Accuracy Evaluation Completed
- Precision & Recall Evaluation Completed
- Confusion Matrix Generated

System Development:

- Flask Backend Completed
- Frontend Integration Completed
- Prediction API Completed
- Grad-CAM Integration Completed

Remaining Functionality

Dashboard Module:

- Patient Statistics
- Analysis Dashboard
- Visualization Charts
- User Management

Database Module:

- Patient Data Storage
- MRI History Storage
- Analysis Records
- Report Management

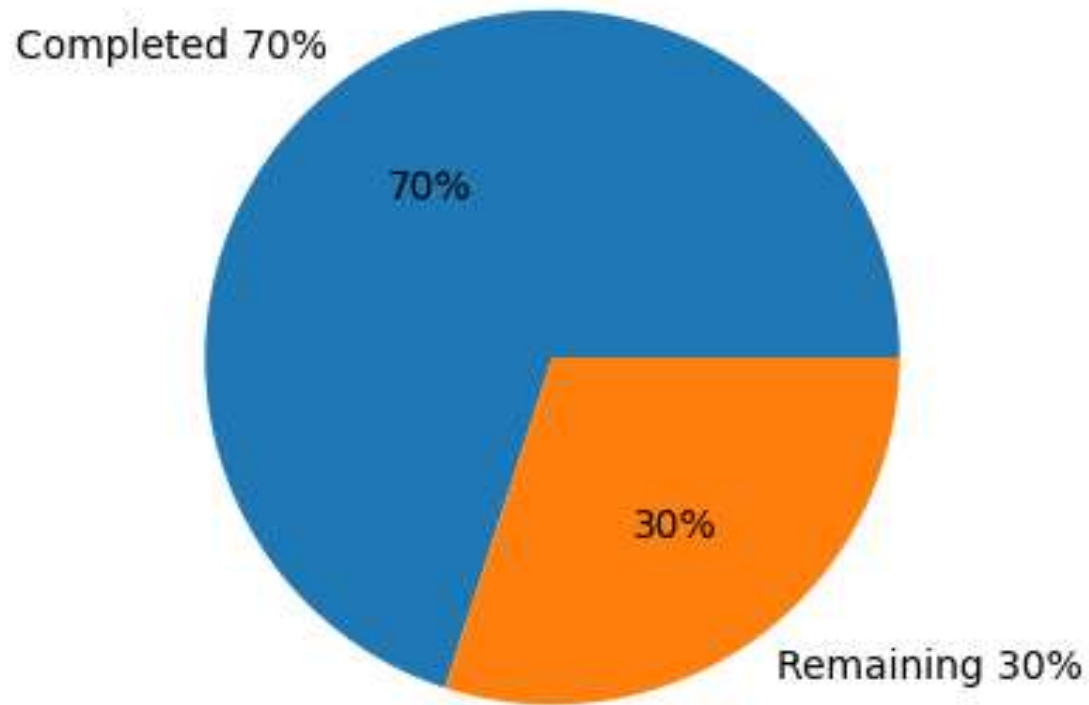
LLM Module:

- Explanation Generation
- Recommendation System
- Automated Report Generation

Final Phase:

- Final Testing
- Deployment
- Optimization
- Documentation

Project Progress



Estimated Completion Time

Task	Estimated Time
Dashboard Development	2-3 days
Database Integration	~1 week
LLM Integration	1 week
Final Testing	2-3 days
Deployment	3-4 days
Documentation	~1 week

Total Remaining Time:

- Approximately ~4 weeks

System UI

 **Alzheimer's Detection System**

AI-Powered | Research Use

1 Patient Details

2 MRI Upload

3 Symptoms

4 Analyze

This tool assists clinical decision-making only. Always confirm findings with a qualified neurologist or radiologist.

Patient Information

Required

FULL NAME

DATE OF BIRTH

AGE

SEX

EDUCATION LEVEL

FAMILY HISTORY

REFERRING PHYSICIAN

SCAN DATE

CLINICAL NOTES

PRELIMINARY RISK FACTORS

MRI Scan Upload

DICOM / PNG / JPG

T1-weighted

T2-weighted

FLAIR

DWI



Upload MRI Scan

Drag & drop or click to browse from your device

Supports .DIPN, .PNG, .BMP, .TIF, and DICOM formats

T1-weighted

Axial view preferred

Hippocampal region

Observed Symptoms

Select all that apply

☐ MEMORY LOSS

☐ DISORIENTATION

☐ LANGUAGE DIFFICULTY

☐ MOOD CHANGES

☐ PROBLEM SOLVING DECLINE

☐ CONFUSION (TIME/PLACE)

☐ PERSONALITY CHANGE

☐ WITHDRAWAL / APATHY

Prediction Result

SCORE	CLASS	CONFIDENCE	QUALITY
100% Very mild dementia indication	VeryMildDemented Probable Stage	99.98% Model Confidence	T1 MR Scan

Symptoms

- Memory loss

Prevention Advice

- Consult a neurologist or radiologist for stage of confirmation.
- Monitor regular cognitive screening and medical follow-up.
- Encourage physical activity, a balanced diet, proper sleep, and mental stimulation.
- Monitor changes in memory, language, mood, orientation, and daily functioning.
- For memory loss, use reminders, calendars, notes, labels, and fixed daily routines.

This results for research and educational purposes only.

Grain-CAM Explainability

Highlighted regions indicate areas that influenced the model prediction.



Project Demonstration

Demonstration Flow:

Step 1: Upload MRI Image

Step 2: Image Preprocessing

Step 3: ResNet50 Prediction

Step 4: Confidence Score Generation

Step 5: Class Probability Display

Step 6: Grad-CAM Heatmap Generation

Step 7: Result Visualization

Best Accuracy: 96.62%